

Improving Wildfire Evacuation Accessibility in California: A Spatial Analysis of Safe Zone

Placement and Travel-Time Reduction

Jesus Perez, Henry Garcia

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Instructor: Vivian Sultan

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Research Question

How can the placement of additional evacuation routes or safe zones reduce the number of high wildfire-risk communities in California unable to evacuate within 30 minutes?

Importance of the Issue

Wildfires in California have become increasingly frequent and severe, posing significant threats to human life and infrastructure. While prior research has examined wildfire risk and evacuation behavior, less attention has been given to identifying communities that may be unable to evacuate within a critical time threshold due to transportation limitations. Understanding where evacuation accessibility is constrained is essential for improving emergency preparedness, reducing loss of life, and guiding infrastructure planning. However, identifying vulnerable areas alone is not sufficient; there is a critical need to evaluate how targeted infrastructure improvements, such as additional evacuation routes or safe zones, can reduce evacuation failure and improve emergency response planning. This study aims to provide a spatially explicit assessment of evacuation vulnerability by integrating wildfire risk data with network-based travel time analysis.

Intended Audience

This research is intended for emergency management agencies, transportation planners, urban and regional policymakers, and public safety officials responsible for disaster preparedness and evacuation planning.

Literature Review and Related Work

Several studies have examined wildfire risk, evacuation behavior, and the application of GIS-based network analysis in disaster scenarios. These studies provide a foundation for

understanding both the spatial distribution of wildfire hazards and the challenges associated with emergency evacuation.

Cova and Church (1997) were among the first to model evacuation vulnerability using GIS, introducing methods to identify communities at risk due to limited road network accessibility. Their work demonstrated that transportation infrastructure plays a critical role in determining whether populations can evacuate efficiently during hazard events. This approach is directly relevant to the present study, which similarly evaluates evacuation capability using road network constraints.

Subsequent research expanded on this foundation by focusing specifically on wildfire scenarios. Cova, Theobald, Norman, and Siebeneck (2013) analyzed wildfire evacuation vulnerability across the western United States, highlighting how limited road connectivity and network bottlenecks can prevent timely evacuation in high-risk areas. Their findings emphasize the importance of integrating transportation network analysis with hazard exposure, which is a central component of this study.

In addition, Church and Sexton (2002) explored evacuation modeling at a finer spatial scale, demonstrating how GIS can be used to estimate whether populations can evacuate within critical time windows. Their work supports the use of travel-time thresholds as a practical way to evaluate evacuation feasibility, reinforcing the methodological approach adopted in this research.

More broadly, GIS-based evacuation planning has been widely recognized as an effective tool for analyzing emergency response scenarios (Chen & Zhan, 2012). These approaches often incorporate service area and network-based travel time analysis to assess accessibility under time constraints, further supporting the use of a threshold-based framework.

While prior studies have successfully identified evacuation vulnerability using network-based approaches, fewer studies evaluate how modifications to infrastructure such as the addition of evacuation routes or safe zones can improve evacuation outcomes. This study extends existing work by incorporating scenario-based analysis to assess how such interventions can reduce evacuation constraints.

Spatial Data Sources

This study will integrate several spatially explicit datasets to analyze wildfire evacuation accessibility in California. The primary data sources include:

- **Wildfire Risk Data:** California Department of Forestry and Fire Protection (CAL FIRE) Fire Hazard Severity Zone (FHSZ) maps, which classify areas based on wildfire susceptibility.
- **Population Data:** U.S. Census Bureau American Community Survey (ACS) data at the census tract or block level to estimate populations exposed to wildfire risk.
- **Road Network Data:** OpenStreetMap (OSM) or U.S. Census TIGER/Line shapefiles, providing detailed road connectivity for network-based travel time analysis.
- **Evacuation / Safe Zone Locations:** California Office of Emergency Services (Cal OES) and local county emergency management agencies.
- **Terrain and Land Cover Data (Optional Enhancement):** U.S. Geological Survey (USGS) datasets, including elevation and slope, to account for geographic factors affecting travel speed and evacuation routes.

These datasets will be integrated within a GIS environment to conduct spatial and network-based analysis. Specifically, the road network will be used to model travel times from population

centers to designated safe zones, while wildfire hazard layers will identify high-risk areas. The analysis will first identify communities that are both exposed to wildfire risk and unable to reach safety within a 30-minute threshold, and will then evaluate how the introduction of additional evacuation routes or safe zones alters these outcomes.

Expected Results / What the Analysis Will Tell You

The analysis is expected to produce both baseline and scenario-based results that evaluate evacuation accessibility under current and improved conditions. Initial outputs will identify communities in high wildfire risk areas that are unable to reach a safe zone within a 30-minute threshold. Improvement will be measured by the reduction in the number of inaccessible census tracts and the percentage decrease in the affected population.

Following the introduction of additional evacuation routes or safe zones, the analysis will generate updated accessibility maps showing how these changes affect travel time and evacuation feasibility.

The findings will be presented through thematic maps, including baseline vulnerability maps, improved accessibility maps, and comparative visualizations highlighting changes between scenarios. Supporting tables and charts will summarize key metrics, allowing for clear evaluation of how different intervention strategies impact evacuation outcomes.

Overall, the results will provide practical insight into how targeted infrastructure improvements can enhance evacuation planning and reduce risk in high wildfire hazard areas.

System

The primary software environment used in this project was ArcGIS Pro, a professional Geographic Information System (GIS) application developed by Esri. ArcGIS Pro was selected because it provides the spatial analysis, network analysis, and cartographic tools necessary for evacuation accessibility modeling. The software supported all major stages of the project, including data preparation, service area analysis, spatial overlay operations, and map visualization.

The project was organized using a file geodatabase structure within ArcGIS Pro to maintain a cleaner and more consistent workflow. Separate feature classes and datasets were created for wildfire hazard zones, census population tracts, road networks, shelter locations, accessibility outputs, and scenario simulation outputs. Organizing the project this way helped reduce processing errors and made the analysis easier to manage throughout development.

The Network Analyst extension in ArcGIS Pro was used to create the transportation network dataset and perform travel-time analysis. The road network was processed into a network dataset capable of supporting routing and service area calculations. Travel impedance was configured using a custom time-based cost field representing estimated driving time in minutes. This allowed the analysis to calculate areas capable of reaching evacuation shelters within a 30-minute threshold.

Several preprocessing steps were required before analysis could begin. Road network data needed attribute cleaning and travel-time preparation because the original dataset did not contain complete travel-time information. Shelter datasets were filtered to California-only facilities and visually checked to confirm that locations aligned reasonably with nearby roads

and populated areas. Census population data from the American Community Survey (ACS) was joined to census tract geometries to support population-based vulnerability analysis.

During implementation, several technical issues were encountered while building and testing the workflow. Some shelter facilities initially failed to connect correctly to the transportation network during service area analysis. Additional problems included empty feature exports, interrupted network analysis sessions, and delays caused by statewide processing. These issues were resolved through repeated validation, rebuilding analysis layers when necessary, correcting exported features, and re-running service area calculations after adjustments were made.

The project also included a scenario simulation component through the creation of hypothetical evacuation safe zones. Four proposed shelter locations were placed near underserved wildfire-risk regions in Patterson, Los Banos, Merced, and Pleasant Valley to test whether accessibility coverage could be improved. These locations were selected based on visible accessibility gaps, limited shelter density, nearby population centers, and proximity to major transportation corridors.

Final outputs generated in ArcGIS Pro included statewide accessibility maps, vulnerable population tract maps, service area polygons, and improved accessibility scenario maps. ArcGIS Pro layout tools were used to create final cartographic figures with legends, thematic symbology, and map elements suitable for inclusion in the final paper and presentation.

Methodology

Data Preparation

The analysis began with the preparation and standardization of multiple spatial datasets within ArcGIS Pro. Wildfire hazard data from CAL FIRE Fire Hazard Severity Zone (FHSZ) maps was filtered to retain only areas classified as “High” and “Very High” wildfire risk. These layers were then combined into a single wildfire-risk feature class representing regions with elevated evacuation vulnerability.

Population data from the American Community Survey (ACS) was joined to California census tract geometries using census tract GEOID identifiers. This process created a tract-level population layer containing both demographic and geographic information necessary for the analysis. The total population field from ACS table B01003 was preserved for later population calculations and vulnerability estimates.

Road network data was also cleaned and prepared before network analysis could be performed. Because the original road dataset did not contain a complete travel-time field, estimated driving times were calculated using road length and assigned speed assumptions. Additional fields representing travel time in minutes were created and populated to support time-based routing analysis.

Shelter-capable facilities were imported from the FEMA National Shelter System Facilities dataset and filtered to California-only locations. Duplicate or invalid shelter points were removed when necessary. Shelter locations were visually inspected to confirm that they aligned reasonably with nearby roads and populated regions before service area analysis was performed.

To maintain spatial consistency throughout the workflow, all datasets were projected into the same coordinate system. The project ultimately used the NAD 1983 California (Teale) Albers projection for statewide analysis and final map production.

Network Dataset Construction

A transportation network dataset was created using the ArcGIS Pro Network Analyst extension. A feature dataset was first created within the project geodatabase to store the cleaned road network and associated network components. The network dataset was then built using the prepared road feature class.

Travel impedance was configured using a custom time-based cost attribute representing estimated driving time in minutes. A custom driving-time travel mode was created using this impedance field. The completed network dataset allowed ArcGIS Pro to perform route calculations, network traversal, and service area accessibility analysis.

The network dataset was validated through several route analysis tests. Sample routes were generated between selected locations to verify that facilities correctly snapped to the road network and that cumulative travel-time calculations were functioning properly. During testing, some facilities initially failed to connect correctly to the network and required additional adjustments before accurate analysis could be completed.

Baseline Evacuation Accessibility Analysis

The baseline analysis focused on identifying wildfire-risk communities unable to access evacuation shelters within a 30-minute driving threshold. Service Area analysis was performed using the Network Analyst extension in ArcGIS Pro.

Existing shelter locations were loaded as facilities into the service area layer, and a 30-minute travel-time cutoff was applied using the custom driving-time travel mode. The Service Area tool generated polygons representing regions capable of reaching evacuation shelters within the specified travel threshold.

The resulting service area polygons were exported and dissolved into a single statewide accessibility surface representing all locations reachable within 30 minutes. Dissolving the polygons simplified overlapping service areas into one continuous accessibility layer suitable for overlay analysis.

To identify inaccessible wildfire-risk regions, the Erase tool was used to remove accessible regions from the statewide wildfire hazard layer. The output represented wildfire-prone areas located outside the 30-minute accessibility threshold.

The inaccessible wildfire-risk layer was then intersected with census population tracts using the Intersect tool. This produced a vulnerable population tract layer representing census tracts simultaneously exposed to elevated wildfire risk and lacking 30-minute shelter accessibility.

Vulnerable Population Analysis

Quantitative population analysis was performed using the Summary Statistics tool within ArcGIS Pro. The ACS total population field (B01003_001E) was summarized across all

vulnerable population tracts to estimate the statewide population exposed to wildfire risk outside the 30-minute accessibility threshold.

The baseline analysis identified approximately 5.4 million residents located within wildfire-risk areas lacking sufficient evacuation accessibility. The analysis demonstrated the existence of major accessibility gaps across portions of California and highlighted the spatial extent of evacuation vulnerability.

Because intersect operations split some census tract geometries into multiple polygons, the resulting feature count reflected intersected spatial records rather than perfectly unique census tracts. Although this introduced the possibility of some duplicated tract representation, the workflow still provided a strong statewide estimate suitable for comparative GIS analysis.

Scenario Simulation and Improvement Analysis

To evaluate potential improvements in evacuation accessibility, a hypothetical shelter expansion scenario was developed. A new point feature class representing proposed evacuation safe zones was created within the project geodatabase.

Four hypothetical shelter locations were manually placed in underserved wildfire-risk regions near Patterson, Los Banos, Merced, and Pleasant Valley. These locations were selected based on visible accessibility gaps, nearby population centers, limited shelter coverage, transportation access, and proximity to wildfire-prone regions.

The proposed shelters were merged with the original shelter dataset to create an improved shelter network layer. The full accessibility workflow was then repeated using the updated shelter configuration. This included rerunning Service Area analysis, dissolving accessibility

polygons, performing erase analysis against wildfire-risk regions, intersecting with population tracts, and recalculating summary statistics.

The improved accessibility maps demonstrated localized expansion of 30-minute evacuation coverage around several previously underserved communities. Although the statewide vulnerable population estimate remained numerically similar to the baseline estimate, the improved scenario illustrated how strategically placed shelters could reduce accessibility gaps and improve evacuation access in targeted regions.

Cartographic Visualization

Final cartographic outputs were created using ArcGIS Pro layout tools. Map layouts included legends, scale bars, north arrows, and thematic symbology representing wildfire hazard severity, shelter locations, proposed shelter additions, and accessibility coverage areas.

Multiple map figures were generated to visualize baseline accessibility, vulnerable population tracts, proposed shelter locations, and improved accessibility scenarios. Transparent accessibility overlays and differentiated point symbology were used to improve readability and clearly distinguish between existing shelters and proposed shelter additions.

Results / Discussion

The results of this study demonstrated the effectiveness of GIS-based network analysis for identifying wildfire evacuation accessibility gaps across California. By combining wildfire hazard zones, census population data, shelter locations, and transportation networks, the analysis identified regions where residents may face difficulty evacuating to safety within a 30-minute travel threshold.

Baseline Accessibility Results

The baseline service area analysis generated statewide accessibility polygons representing areas capable of reaching evacuation shelters within 30 minutes of driving time. Accessibility coverage was strongest in densely populated urban and suburban regions where shelter density and transportation connectivity were higher. In contrast, significant accessibility gaps appeared in rural and mountainous wildfire-prone regions where shelter coverage was sparse and road connectivity was more limited.

The overlay analysis between wildfire hazard zones and the statewide accessibility surface identified multiple high-risk regions located outside the evacuation threshold. These inaccessible areas were concentrated primarily along rural wildfire corridors and regions with lower infrastructure density. The resulting inaccessible wildfire-risk layer highlighted communities that may experience greater evacuation difficulty during emergency wildfire events.

The inaccessible wildfire-risk layer was then intersected with census population tracts to identify communities simultaneously exposed to elevated wildfire risk and lacking sufficient evacuation accessibility. Summary statistics calculated from the ACS total population field estimated that approximately 5.4 million residents were located within wildfire-risk areas outside the 30-minute evacuation threshold.

The baseline analysis demonstrated that wildfire evacuation vulnerability is not limited to isolated areas, but instead affects large portions of California. The results also illustrated how shelter distribution and transportation infrastructure strongly influence evacuation accessibility outcomes across the state.

Proposed Shelter Scenario Results

To evaluate whether additional shelter infrastructure could improve accessibility conditions, a scenario simulation was developed using four hypothetical evacuation safe zones placed near underserved wildfire-risk regions. Proposed shelter locations were added near Patterson, Los Banos, Merced, and Pleasant Valley. These locations were selected based on visible accessibility gaps, nearby population centers, limited shelter coverage, and transportation access.

The improved shelter scenario was processed using the same network analysis workflow as the baseline analysis. New service area polygons were generated and dissolved to create an updated accessibility surface. The resulting maps showed localized improvements in accessibility coverage near the proposed shelters. Areas surrounding the proposed facilities displayed expanded 30-minute accessibility zones extending toward previously underserved wildfire-risk regions.

Visual comparison between the baseline and improved accessibility maps suggested that strategically placed shelters can improve evacuation accessibility for targeted communities. The proposed shelter locations helped reduce several local accessibility gaps and expanded service coverage near transportation corridors and vulnerable population clusters.

However, summary statistics calculated from the improved scenario produced a statewide vulnerable population estimate similar to the baseline result. This outcome suggests that wildfire evacuation vulnerability in California remains geographically widespread and may not be substantially reduced through a limited number of shelter additions alone.

Although the numerical reduction in vulnerable population was limited, the scenario analysis still demonstrated the value of targeted infrastructure planning. The proposed shelters improved accessibility conditions in localized areas and illustrated how GIS-based modeling can support emergency evacuation planning and infrastructure decision-making.

Interpretation of Findings

The findings of this study reinforce the importance of transportation accessibility in wildfire evacuation planning. Communities located farther from shelters or connected by limited transportation infrastructure were more likely to fall outside the evacuation threshold. This pattern was especially visible in rural and mountainous wildfire-prone regions.

The scenario simulation also demonstrated that localized accessibility improvements do not always translate into major statewide population reductions. California's wildfire exposure is geographically extensive, and evacuation accessibility is influenced by both shelter distribution and broader transportation connectivity. These findings suggest that evacuation planning may require larger-scale infrastructure investments, expanded shelter networks, or additional transportation improvements beyond a small number of shelter additions.

Despite these limitations, the project successfully demonstrated how GIS network analysis can identify evacuation accessibility gaps, estimate vulnerable populations, and evaluate potential infrastructure improvements. The workflow developed in this study provides a reproducible framework for wildfire evacuation accessibility analysis and supports the use of GIS as a decision-support tool for emergency management and spatial planning.

The project also demonstrated the practical value of combining multiple GIS techniques, including service area analysis, network modeling, spatial overlay analysis, and population-based vulnerability analysis. The resulting maps and quantitative outputs provide a clearer spatial

understanding of evacuation accessibility conditions and can support future planning and emergency preparedness efforts.

Limitations

This project contains several limitations related to data quality, modeling assumptions, and computational constraints. First, the wildfire hazard, shelter, and road network datasets may contain incomplete or outdated information that could affect accessibility calculations. Shelter locations were treated as fully operational evacuation destinations, even though actual shelter availability may vary during real emergency situations.

The transportation network analysis also relied on simplified travel assumptions. Real-world evacuation conditions such as traffic congestion, road closures, wildfire spread, weather conditions, and road capacity limitations were not included in the model. As a result, actual evacuation travel times during a wildfire event could differ significantly from the estimated 30-minute accessibility results produced in this study.

In addition, the intersect analysis used census tract population totals, which may introduce some population overcounting because tract geometries were split into multiple intersected features during overlay analysis. While the workflow still provides a strong statewide estimate of vulnerable populations, more advanced methods such as area-weighted population analysis could improve accuracy in future studies.

Another limitation is that the scenario simulation included only four hypothetical shelter additions. Although these proposed shelters improved localized accessibility coverage, larger-scale infrastructure changes or a broader shelter expansion strategy would likely be necessary to produce more substantial statewide reductions in evacuation vulnerability.

Finally, the project focused primarily on accessibility and travel-time analysis rather than real-time evacuation behavior. Human decision-making, emergency response coordination, and changing wildfire conditions were outside the scope of this study, but could significantly influence evacuation outcomes in real-world scenarios.

Conclusions

This project demonstrated how GIS-based network analysis can be used to evaluate wildfire evacuation accessibility across California. By integrating wildfire hazard zones, transportation networks, shelter locations, and census population data, the study identified communities located in wildfire-risk areas that may be unable to evacuate to safety within a 30-minute travel threshold.

The baseline analysis revealed that wildfire evacuation vulnerability affects large portions of California, particularly in rural and mountainous regions with limited shelter coverage and transportation connectivity. The results estimated that approximately 5.4 million residents were located within wildfire-risk areas outside the 30-minute evacuation accessibility threshold. These findings highlight the importance of evacuation infrastructure and accessibility planning in reducing wildfire-related risk.

The project also evaluated a hypothetical improvement scenario through the addition of strategically placed evacuation safe zones near Patterson, Los Banos, Merced, and Pleasant Valley. The improved scenario demonstrated localized expansion of accessibility coverage and illustrated how additional shelters can improve evacuation access in underserved regions.

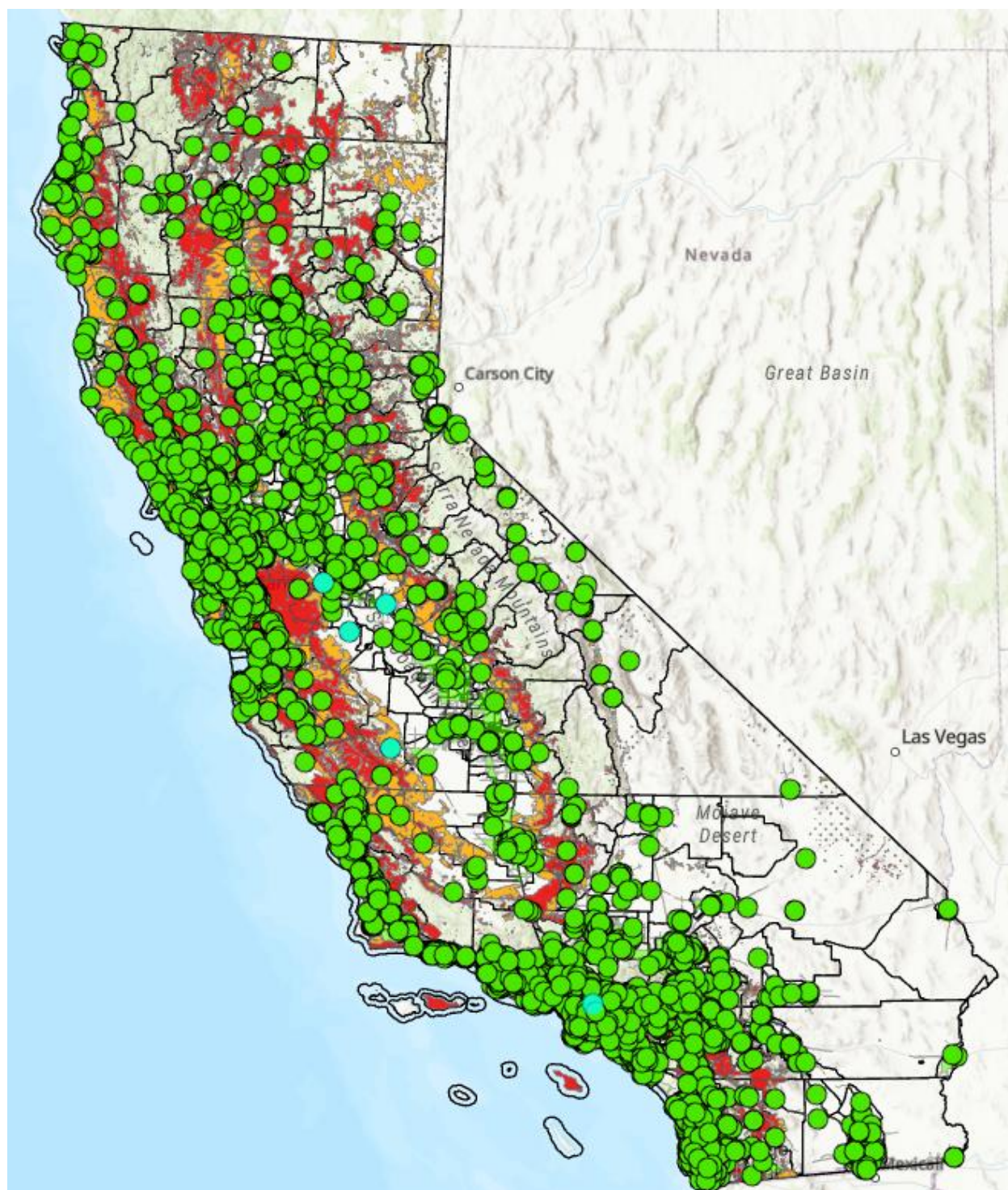
Although the statewide vulnerable population estimate did not significantly decrease, the scenario analysis showed that targeted infrastructure improvements can still provide meaningful local accessibility benefits.

Overall, this study demonstrated the value of GIS as a decision-support tool for emergency management and evacuation planning. The combination of service area analysis, network modeling, and spatial overlay analysis provided a practical framework for identifying accessibility gaps and evaluating potential infrastructure improvements. The project also showed how GIS can be used not only to identify vulnerability, but also to test possible solutions through scenario-based spatial analysis.

Future research could improve the model by incorporating dynamic wildfire spread, traffic congestion, road capacity limitations, and more advanced population allocation methods to better represent real-world evacuation conditions. Additional research could also evaluate larger-scale shelter expansion strategies or alternative evacuation route planning approaches to further improve statewide accessibility coverage.

Figures/Maps

Figure 1. Overall Map



Legend

-  Safe zone
-  Proposed Safe Zone
-  High
-  Very High

Figure 2. Proposed evacuation safe zone near Patterson.

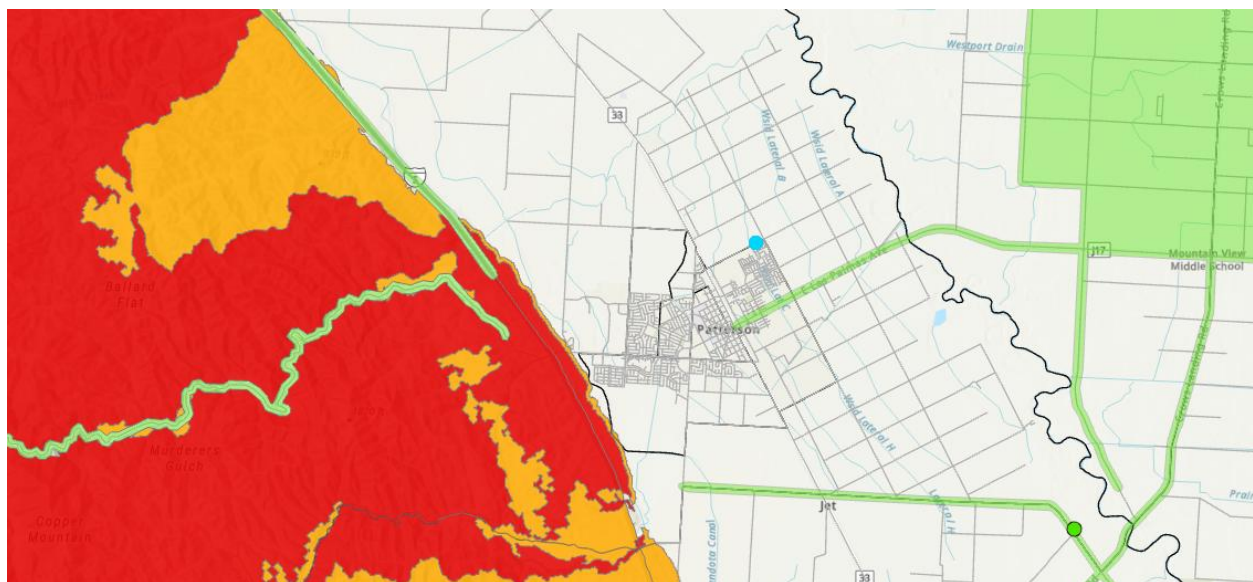


Figure 3. Proposed evacuation safe zone near Los Banos.

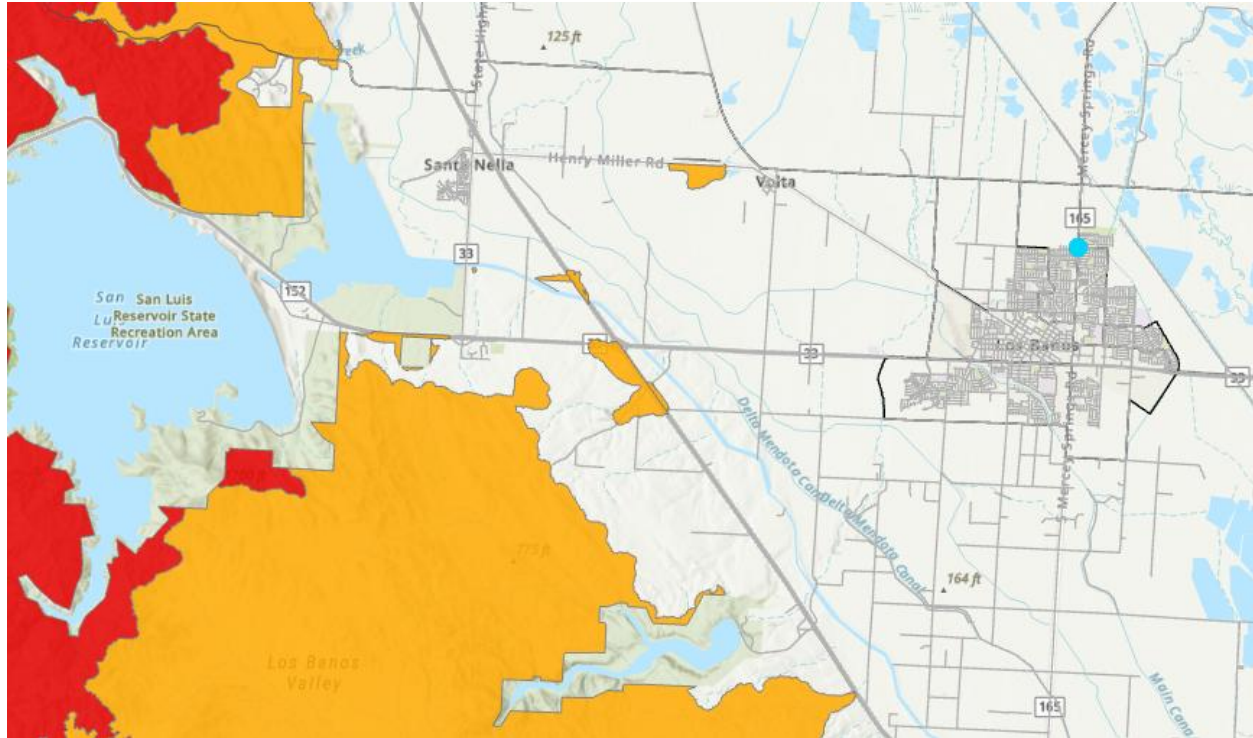


Figure 4. Proposed evacuation safe zone near Merced.

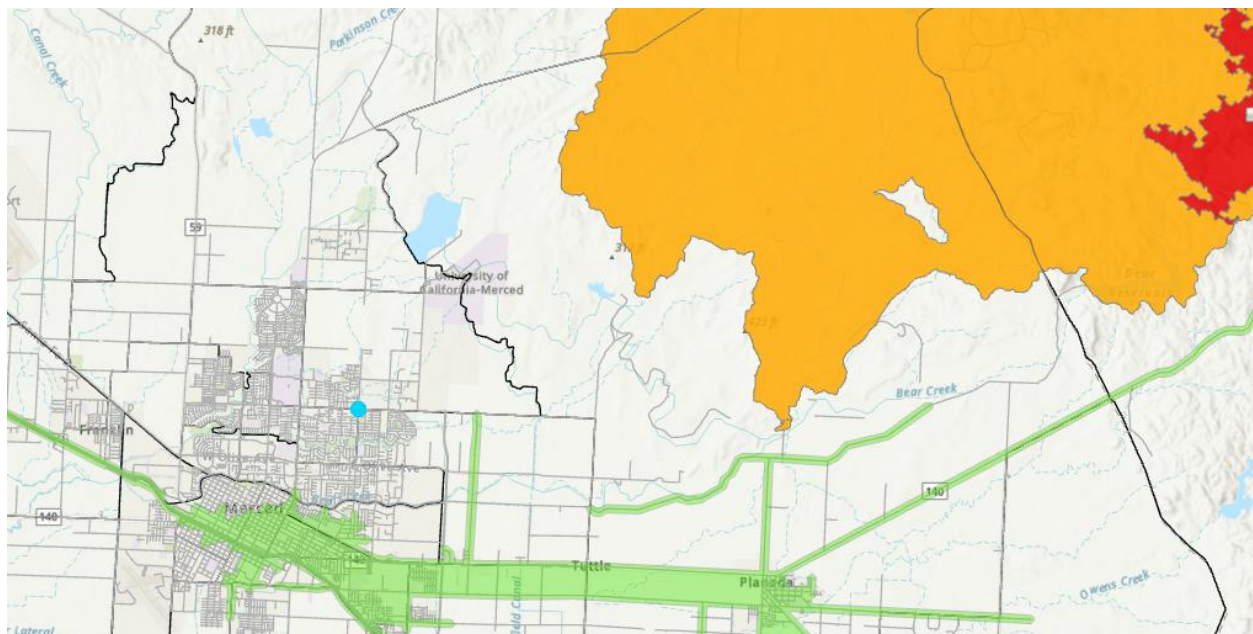


Figure 5. Proposed evacuation safe zone near Pleasant Valley.

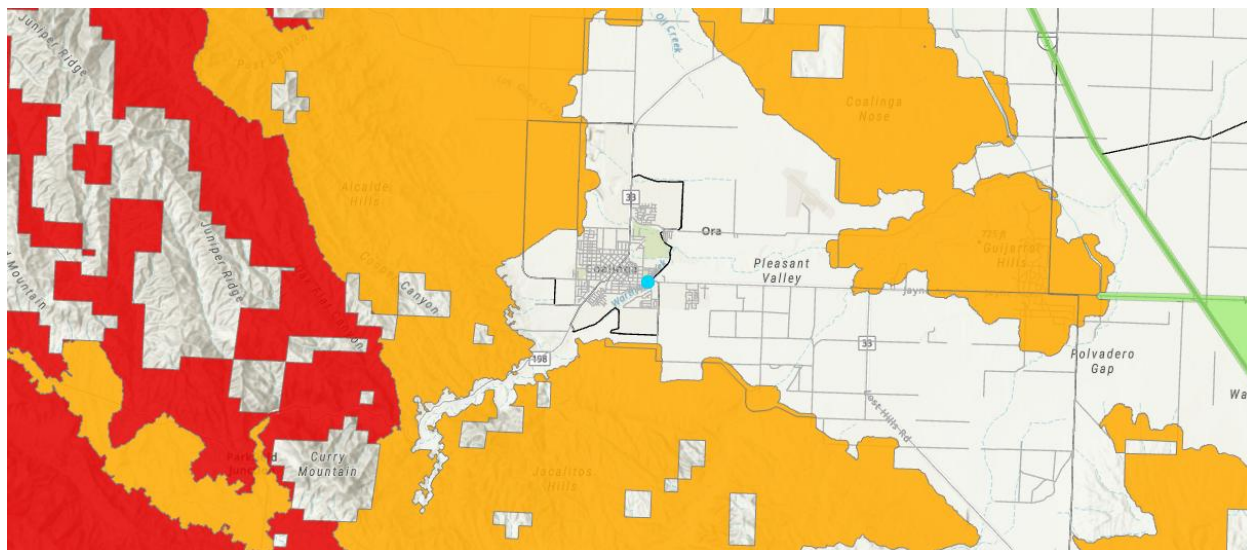


Figure 6. Improved evacuation accessibility coverage near Patterson after the addition of a proposed safe zone.

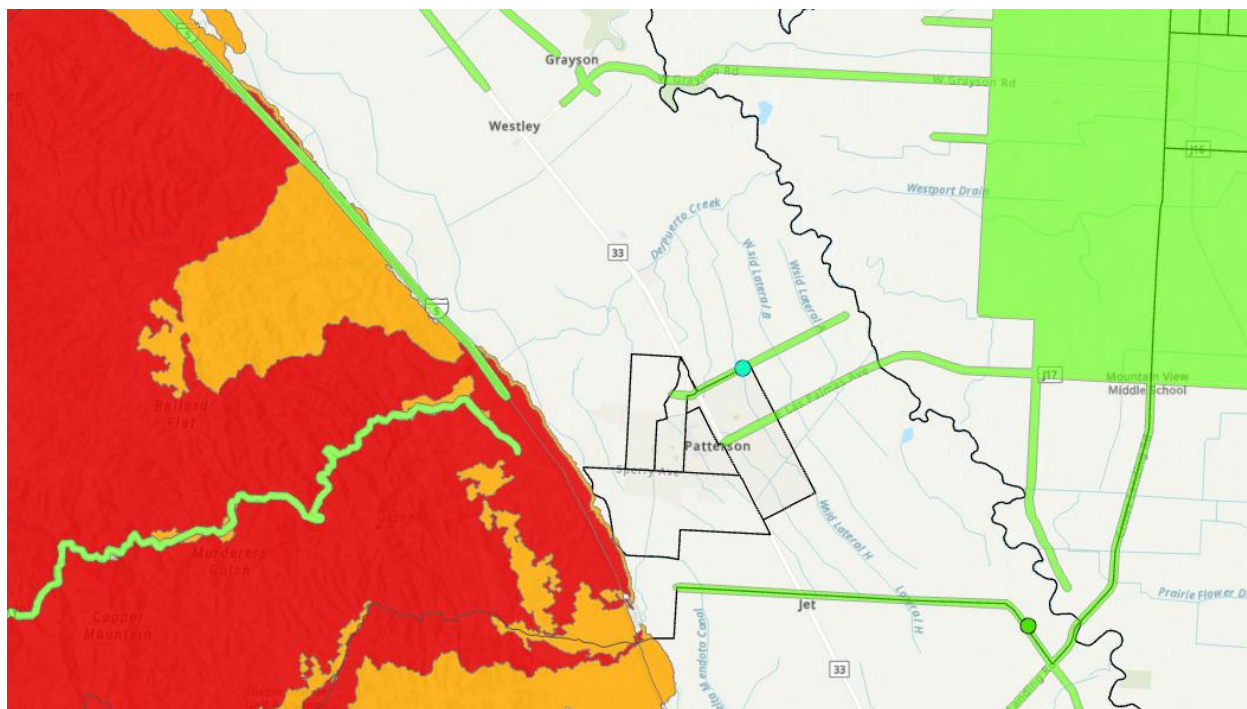


Figure 7. Improved evacuation accessibility coverage near Los Banos after the addition of a proposed safe zone.

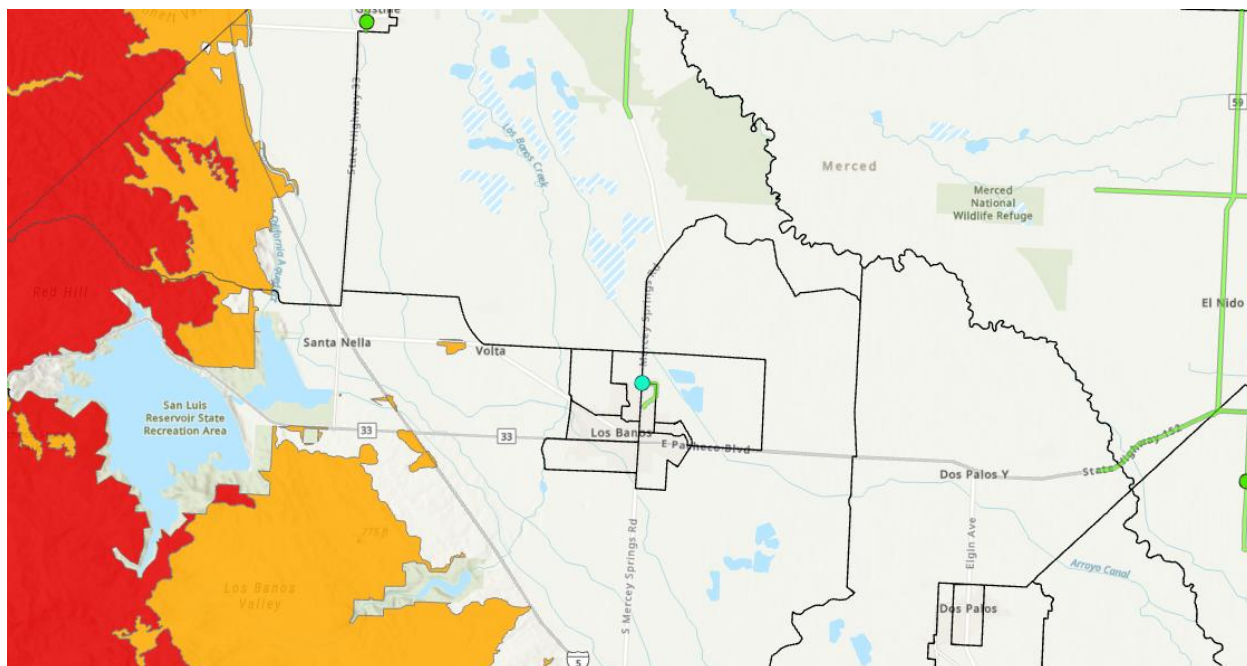


Figure 8. Improved evacuation accessibility coverage near Merced after the addition of a proposed safe zone.

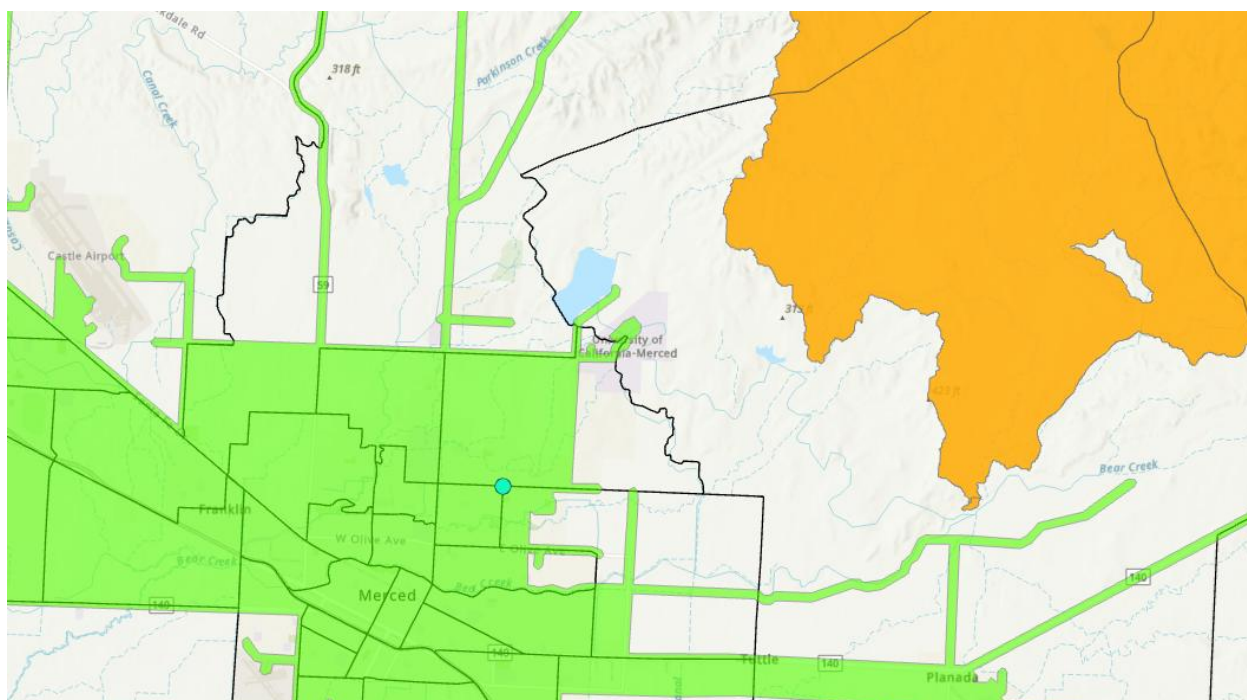
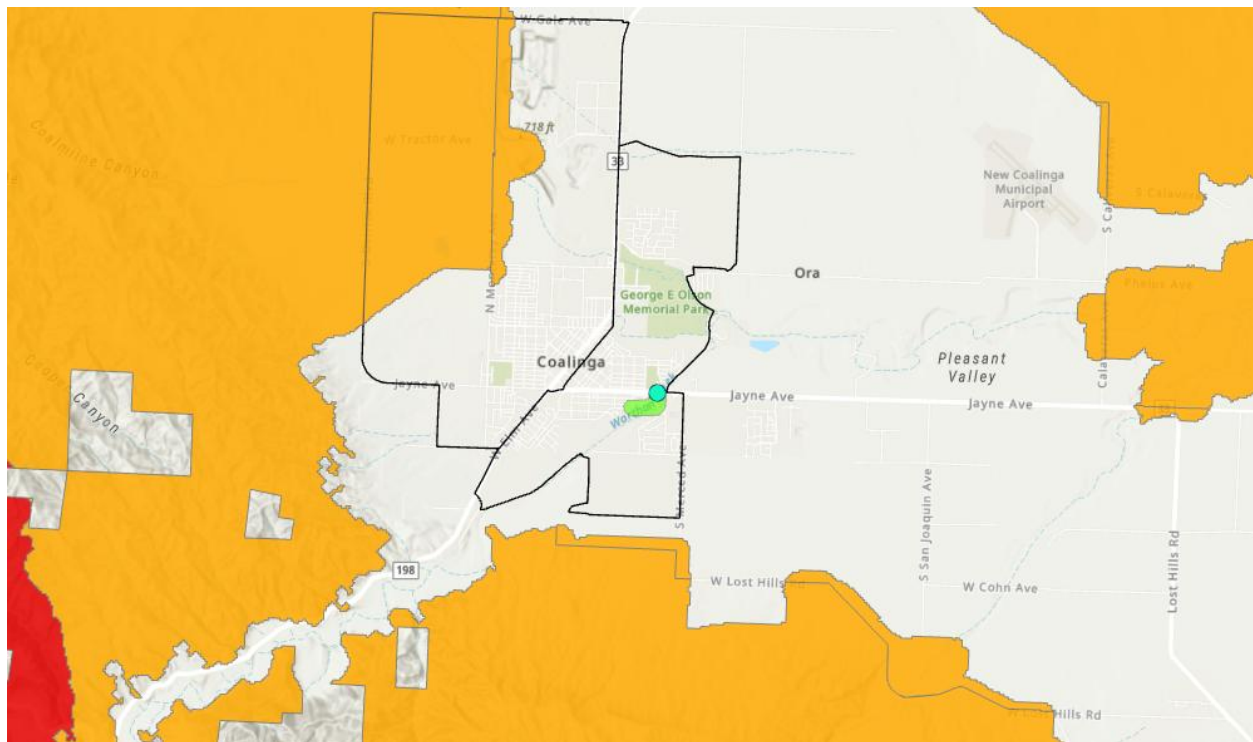


Figure 9. Improved evacuation accessibility coverage near Pleasant Valley after the addition of a proposed safe zone.



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